

Readscape

A Personal Reading Intelligence System





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Research Question

How can reading history data be used to generate personalized book recommendations while also providing readers with meaningful insights about reading habits?



What is Readscape?

- ❖ Readers can use Readscape to build a personal digital library, categorize books as Want to Read, Currently Reading, or Finished, track reading progress, view personalized reading statistics, receive book recommendations.
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Why we chose it?

- ❖ Readers often struggle to keep track of the books they want to read, are currently reading, and have already finished. Their reading information may be scattered across notes, apps, screenshots, or physical lists, making it difficult to organize their library, and understand their reading habits.
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Why it's Important?

Socioeconomic Gap



“Most adults in the study with low literacy levels did have full-time employment, but their work was primarily confined to the service, labor, and construction industries... they often contain few opportunities for professional growth.”

- Maren Blanchard, *The Relationship between Socioeconomic Status and Literacy: How Literacy is Influenced by and Influences SES*

Literacy Crisis



“On average, 79% of U.S. adults nationwide are literate in 2024.”

“54% of adults have a literacy below a 6th-grade level (20% are below 5th-grade level).”

- National Literacy Institute

Community Building



“Stories also have implications for our social world, potentially helping us to better understand other people. While engaging with stories we imagine the experiences of others—including those quite different from ourselves—and doing so may help us hone valuable social cognitive processes (e.g., empathy, perspective-taking).”

- *The Benefits of Reading Narrative Fiction*

User Breakdown



Readers

- ❖ Primary user of Readscape. Can discover track, share, and engage with books and reading communities.



Moderators

- ❖ Monitor community content, manage reports, and enforce community guidelines.



System Administrators

- ❖ Manage users, permissions, system configurations, and overall platform operations



Readers will be able to...

- Create an account and login using secure password or use SSO.
 - Search for books in a user friendly interface.
 - Add books to their personal library.
 - Change a book's reading status.
 - Add friends
 - Record progress, ratings, and completion dates
 - Set reading goals and view analytics
 - Remove books from their library
 - View their reading history
 - Receive recommendations
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Moderators will be able to...

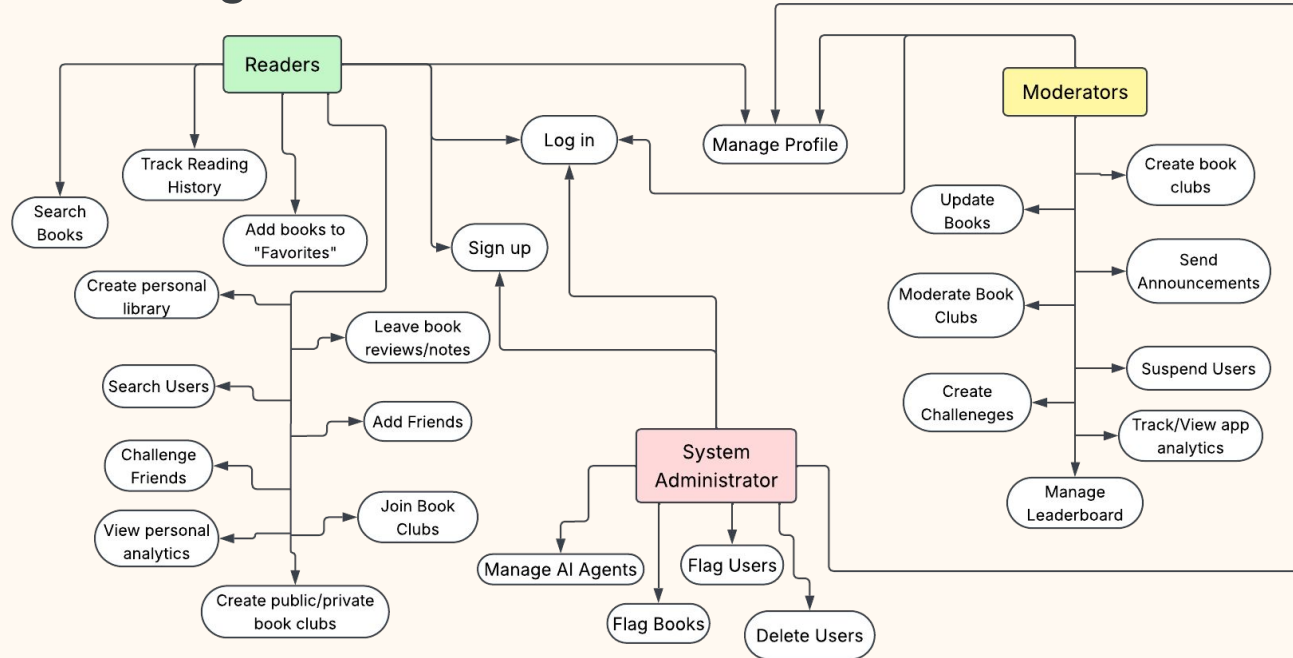
- Retrieve book information using a book API
- Calculate statistics from reading history
- Display statistics through charts and dashboards
- Generate recommendations based on reading preferences and history
- Store user and book information in a database
- Create challenges for users
- Create and moderate book clubs
- Send out updates and announcements
- Suspend Users



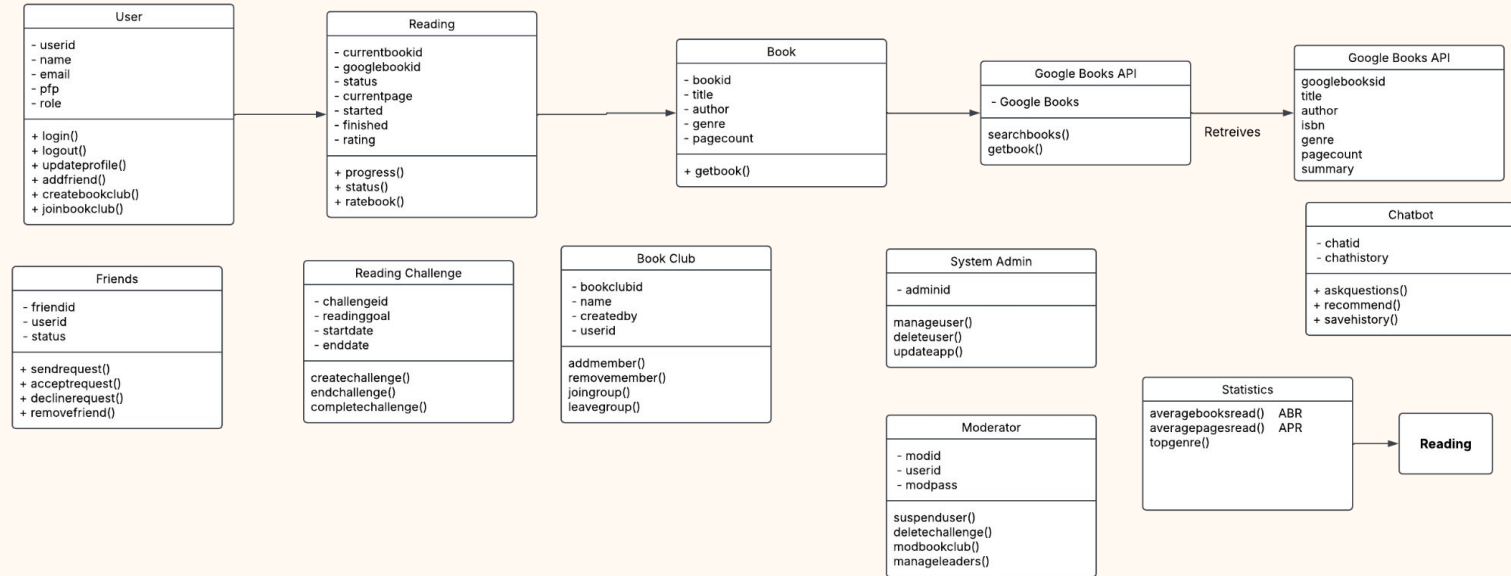
System Admins will be able to...

- Flag and remove books from app
 - Manage AI agents
 - Flag and remove users
 - Manage UI/UX
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Use-Case Diagram



UML Diagram



Project Requirements

- Full Stack Development
 - API Integration
 - Database Management
 - Explainable AI Model
 - Persistent AI Agent
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Specifications

This project will focus on personal reading and data analytics using Spring Boot Framework, Postgresql, and Neon Database. Readscape will be both a mobile and web application.

Tech Stack

SQL

Database Management

React

CSS Framework

**Spring Boot
& Security**

**Google
Books API**

Book Library

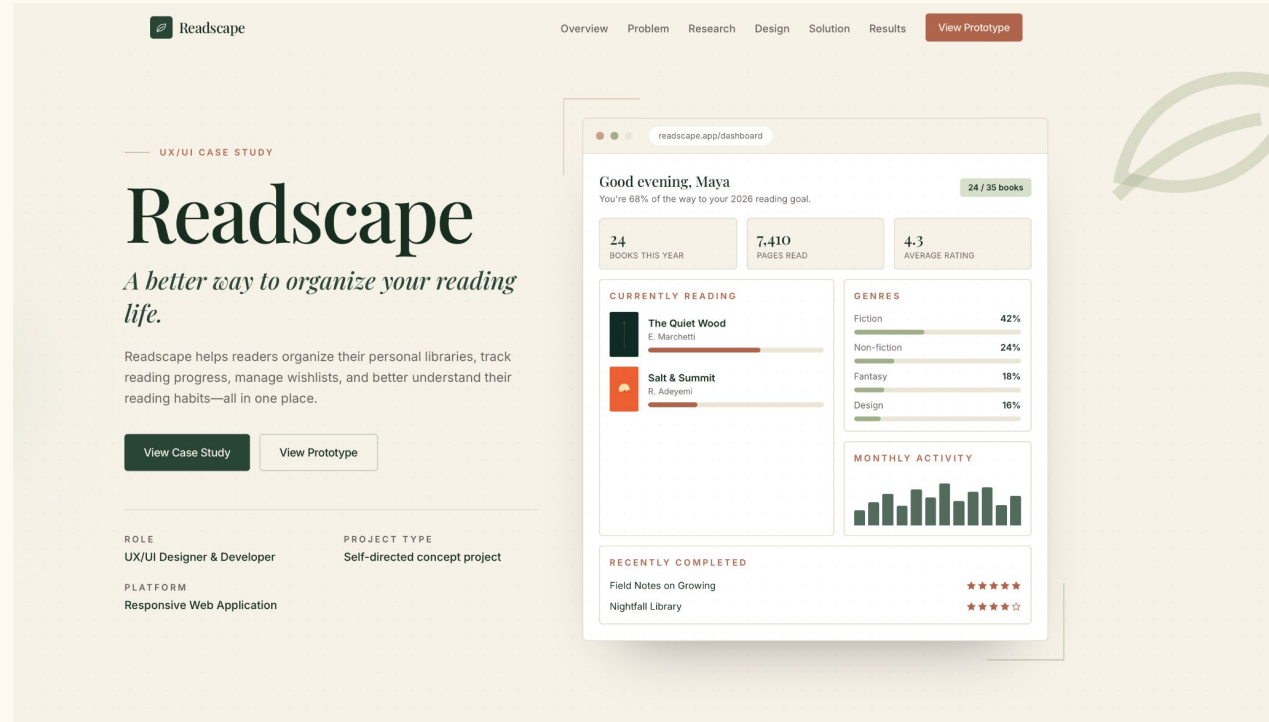
**Google Maps
API**

Locations services

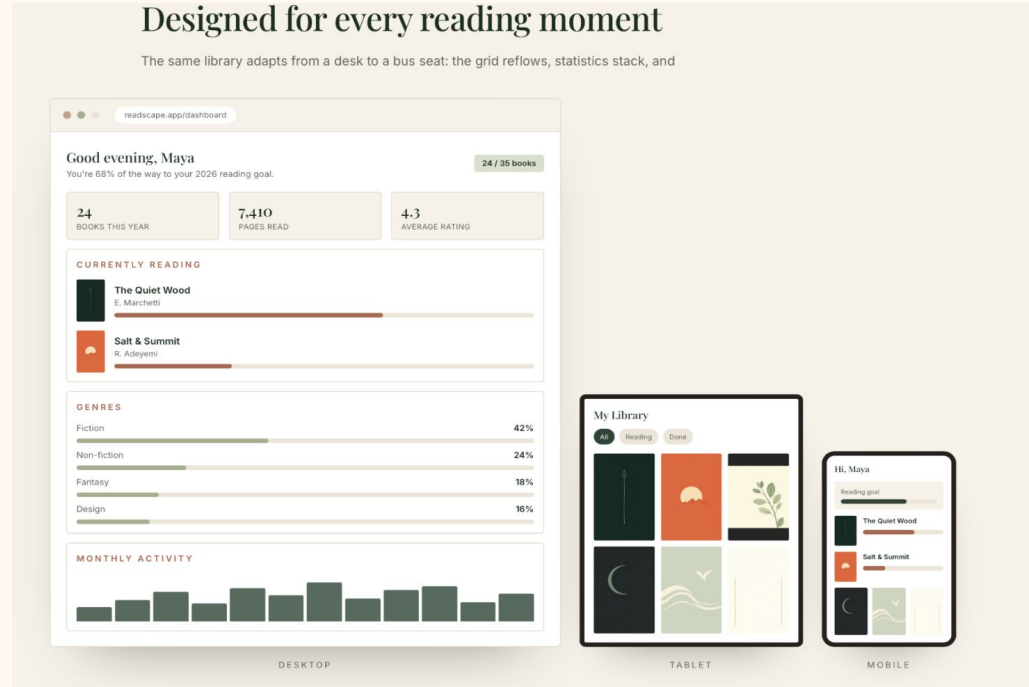
**Google
Oauth 2.0**

Single-Sign On

Prototype



Prototype



Potential Capabilities

Book Cover Recognition & Info Retrieval

- ❖ Camera-based book identification and metadata retrieval with options to save, share, or add the book to a reading list

Geolocation-based Reading Discovery

- ❖ Location-aware discovery of nearby readers, book clubs, events, and challenges

University-Affiliated Community & Collaboration

- ❖ University affiliation enabling users to create, discover, and participate in specific communities

Competitive Reading System

- ❖ Friends can participate in reading competitions using metrics such as books completed, pages read, or reading progress

Peer Reading Accountability System

- ❖ Users can designate a friend as an accountability partner to share progress, encourage reading consistency, and support reading goals

AI-Powered Reading Assistant

- ❖ Provides assistance for book-related questions, author information, personalized recommendations, and discussions
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Teamwork Division

❖ Hiwete

Backend
Development, API
Implementation,
Database
Integration and
Management

❖ Huda

Frontend
Development

❖ Jasmine

AI Integration,
Application
Security & Data
Protection



Thank You !

Questions?
